

An Analytical Failure Criterion for Per-Site Dynamic Operating Envelopes under Mobile Shared Flexible Loads

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Abstract—Dynamic operating envelopes (DOEs) allocate a connection-specific, per-interval power limit to each flexible customer and are secure by construction in the linearised network model used to compute them. This paper asks when that guarantee survives contact with a fleet of electric buses that moves between charging hubs on the same feeder, and answers it with a closed-form criterion. Two error terms are derived from the branch-flow equations and shown to be additive at first order: the loss term the linearised model omits, which is quadratic in power and carries the squared shared-resistance fraction; and the drop caused by real dispatch moving within a 15-minute envelope interval, which is linear in the intra-interval drift and carries the shared-resistance fraction to the first power. Their sum, normalised by the planning band, gives a single inequality computable from topology and fleet parameters alone. Three supporting results are established: corner certification of an envelope box remains exact at N hubs; the loss term admits an exact N -hub quadratic form whose coupling matrix is a Gram matrix and hence positive semidefinite; and the mobility constant does not enter the voltage criterion, so the mobile case reduces to the stationary distributed-energy-resource condition. Validation uses 4760 static allocations on 62 topologies and 48 full-day closed-loop simulations (69 120 minutes) on the IEEE 33-bus and 69-bus systems and two purpose-built synthetic feeders, with every reported violation read from a backward/forward-sweep AC solver rather than from the model the allocation was optimised against. Adding the aggregation term raises per-minute recall from 5.6% to 82.2% with no false positives; on 28 configurations held out from threshold selection, precision and false-positive rate are preserved exactly while recall is 0.852–1.000. A decomposition shows the fleet drift magnitude is invariant across every run and the entire coupling dependence is the network sensitivity weight, matching measurement to three significant figures on four topologies with no fitted quantity. A manipulation experiment isolates the resistance profile of the hub's path as a driver of violation incidence but not of the drift mechanism; the sign of the certified-slack derivative on the 69-bus system remains unresolved and is scoped as future work.

Index Terms—Dynamic operating envelopes, electric bus charging, distribution network security, LinDistFlow, flexibility allocation, hosting capacity.

NOMENCLATURE

u_i	Squared voltage magnitude at bus i [p.u.]
V_{ref}	Substation voltage set point [p.u.]
V_{min}	Statutory lower voltage limit [p.u.]
V_{plan}	De-rated planning floor, $V_{\text{min}} + \text{band}$
R_{ij}, X_{ij}	Resistance/reactance shared by the root-to- i and root-to- j paths [p.u.]
$R_{m\ell}$	Resistance shared by bus m and branch ℓ
r_ℓ	Resistance of branch ℓ [p.u.]
ρ	Shared-resistance fraction R_{AC}/R_{AA}
H	Number of charging hubs
$P_h, \bar{P}_h$	Power and envelope cap of hub h [kW]
S_ℓ, B_ℓ	Total and base apparent flow in branch ℓ
L_k	Loss flowing through branch k
$\mathcal{P}(h)$	Root-to- h branch path
$C^{(m)}, b^{(m)}, c^{(m)}$	Coefficients of the N -hub quadratic form
S_b	Power base [kVA]
Δ	Envelope interval length (15 min)
δp_j	Intra-interval injection deviation at bus j
γ_j, γ_C	Intra-interval drift bound and its measured value at hub C [kW]
w_C	Voltage sensitivity to hub- C power
$D_{\text{err}}, D_{\text{agg}}$	Loss and aggregation error terms [p.u. of u]
Λ_{tot}	Total failure index
τ	Mobility constant (travel plus dwell time)
α	Fairness exponent of the allocation objective

I. INTRODUCTION

DYNAMIC operating envelopes are becoming the standard mechanism by which distribution network service providers expose spare network capacity to flexible customers [1], [2]. A DOE is, by definition, connection-specific and per-interval: for each connection point and each market interval the operator publishes an admissible power range, and any dispatch inside the published set is guaranteed not to breach a network constraint. The guarantee is delivered by solving an allocation problem over a linearised network model, most often the LinDistFlow branch-flow relaxation [3]–[5], and it is exact *within that model*.

Electric bus fleets stress two of the assumptions behind that construction at once. First, several opportunity-charging hubs commonly sit on the same feeder, often on the same lateral, so their envelopes are not independent: the network resource being allocated is shared. Second, and less obviously, the load itself is mobile. A vehicle that charges at one hub departs and reappears at another [6], so charging demand is not a property of a connection point but of a fleet whose spatial distribution changes on a timescale comparable to the envelope interval.

The practical question is not whether a DOE is correct—it is, in its own model—but under what conditions the residual between that model and the physical network exceeds the safety margin the operator has retained. This paper answers that question analytically, and the answer has two components that have not previously been separated. One is the familiar linearisation residual. The other, which dominates in closed

loop, is a *temporal aggregation* residual: the envelope is computed once per interval against interval-mean injections, while dispatch and exogenous load move within the interval, and that movement propagates to a critical bus through the resistance the hubs share.

A. Contributions

- 1) An exact N -hub expression for the loss term omitted by LinDistFlow, obtained by an exchange of summation, and its expansion into a quadratic form whose coupling matrix is a Gram matrix and therefore positive semidefinite (Theorem 2). Corner certification of an envelope box is shown to remain exact at N hubs (Theorem 1).
- 2) A first-order temporal-aggregation term in the same units and the same certification framework, so the two are additive, together with a planning-time bound expressed in drift rates and interval length (Section IV-C). The resulting single inequality (10) is computable from topology, base-case flows and allocated caps without simulation.
- 3) A demonstration that the mobility constant τ does not enter the voltage criterion—the constraint is instantaneous and electrical—and that the criterion reduces to the stationary single-site condition as $\tau \rightarrow 0$ (Section IV-E).
- 4) A validation protocol in which no reported violation is read from the model an allocation was optimised against, spanning 4760 static allocations on 62 topologies and 48 full-day 1-minute closed-loop simulations on four topologies, with two published test feeders held out from the derivation set.
- 5) A decomposition establishing that the entire coupling dependence of the aggregation term is the network sensitivity weight rather than any change in fleet behaviour, giving the closed form (13), and a manipulation experiment that separates what controls the mechanism from what controls violation incidence (Section VII).

B. Organisation

Section II positions the work. Section III states the network and fleet model and the assumptions. Section IV derives the two error terms and the criterion. Section V gives the validation protocol. Section VI reports results, including the holdout and second-topology tests. Section VII decomposes and manipulates the mechanism. Section VIII generalises beyond bus fleets, Section IX states threats to validity, and Section X concludes.

II. RELATED WORK

Work on operating envelopes divides broadly into allocation methods—how to share a feeder’s headroom fairly and efficiently among connection points—and deployment studies that report what happens when envelopes meet real customers. The allocation literature is where the α -fair family used as the benchmark in this paper originates [7], and it has been developed for envelopes in both single-phase and unbalanced

settings, with explicit treatment of the equity of the resulting allocation [8], [9]. Its defining commitment is that limits are connection-specific rather than portfolio-wide [1]. Robust formulations extend the same construction to hedge against forecast and data uncertainty [2].

Deployment trials have independently reported that envelope accuracy is sensitive to how often the allocation is refreshed against updated forecasts, with close-to-real-time envelopes outperforming day-ahead and intra-day ones [10]. The present work explains why refresh cadence matters in a way that is inseparable from network coupling: the error a stale envelope admits reaches a critical bus through shared resistance, so cadence alone does not determine whether a violation occurs.

Charging-coordination work for electric bus fleets has largely treated the network either as a single aggregate constraint or through full nonlinear optimal power flow; scheduling formulations in the transport literature optimise charging against cost and grid impact without a per-connection network limit [6]. The intermediate regime this paper addresses—an operator publishing per-site limits, with the fleet optimising inside them—is the one that matches how DOEs are actually deployed, and it is where the two error terms below become visible.

III. SYSTEM MODEL AND ASSUMPTIONS

A. Network

The feeder is radial with a single feed at bus 0 held at V_{ref} . Write $u_i = |V_i|^2$. Under the LinDistFlow relaxation [3],

$$u_i = V_{\text{ref}}^2 - 2 \sum_j (R_{ij} p_j + X_{ij} q_j), \quad (1)$$

where R_{ij} is the total resistance of the branches common to the root-to- i and root-to- j paths. Because R_{ij} is a sum of positive branch resistances, $R \succeq 0$ elementwise—the property that makes the corner argument of Section IV-A valid.

Assumption 1. *Radial network, single feed held at V_{ref} .*

Assumption 2. *Voltages follow (1) at the planning stage.*

Assumption 3. *Reactive injections are exogenous; hubs exercise no reactive control.*

Assumption 4. *Losses are small relative to flows, so a single loss-correction step suffices.*

Assumption 2 applies only at the planning stage; evaluation always uses the exact AC solution, and Assumption 4 is what licenses the single correction step of (4). Fig. 1 shows the placement geometry used throughout: hub A sits at a lateral tip, hub C is relocated along the path to it so that the shared-resistance fraction $\rho = R_{AC}/R_{AA}$ is swept while the base case is left unchanged, and hubs B and D sit on separate laterals.

B. Envelope allocation

For each interval the operator solves, over the hub caps $P \in \mathbb{R}_+^H$,

$$\max_P \sum_h f_\alpha(P_h) \quad \text{s.t.} \quad u(P) \geq V_{\text{plan}}^2, \quad P \leq \bar{P}^{\text{PCC}}, \quad (2)$$

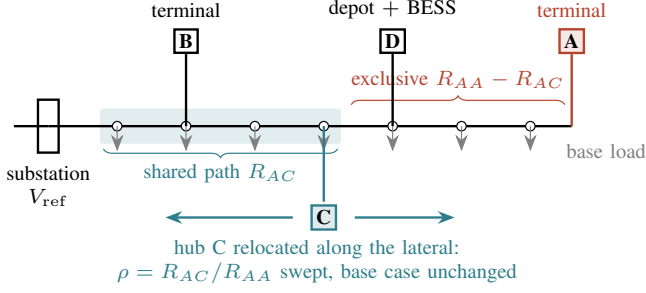


Fig. 1. Coupling geometry. Hub C is a bare en-route charging point with no site load and no photovoltaics, so relocating it along hub A's path changes ρ without changing the base case.

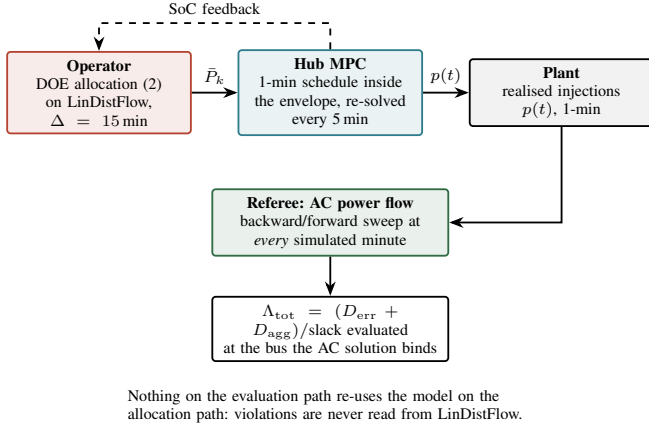


Fig. 2. Control architecture and the evaluation discipline. Every voltage reported as a violation is computed by an AC power flow on the realised 1-minute injections, never by (1).

with f_α from the α -fair family [7] ($\alpha = 0$ efficiency, $\alpha \rightarrow \infty$ max-min). Problem (2) is a *joint* allocation: any combination inside the resulting box is secure in the linear model. It is distinguished throughout from the *one-at-a-time* construction, in which each hub's headroom is computed with the others held at base—a box that is not jointly feasible, and that is included here only because a practitioner reading sensitivity factors off a load-flow tool will produce it.

C. Fleet and control architecture

The lower level is a 1-minute model-predictive controller per hub, re-solved every five minutes, that schedules charging inside the published envelope subject to vehicle state of charge, bay availability and departure times. Traction energy is exogenous: it is computed from longitudinal vehicle dynamics over a synthetic urban duty cycle. The mobility constant τ denotes the time for dispatch at one hub to affect state at another, i.e. travel plus dwell. Fig. 2 shows the architecture and, in particular, the separation between the allocation path and the evaluation path.

IV. DERIVATION

A. Corner certification at N hubs

Theorem 1. Under Assumptions 1–3, $\min_i u_i$ over the box $\prod_h [0, \bar{P}_h]$ is attained at the maximal corner $P = \bar{P}$, for any

number of hubs H .

Proof. From (1), $u_i(P) = u_i(0) - 2 \sum_h R_{ih} P_h$ with $R_{ih} \geq 0$, so each u_i is non-increasing in every coordinate of P . A function non-increasing in each coordinate attains its minimum over a box at the maximal corner; the pointwise minimum over i of such functions inherits the property. $\square$

Theorem 1 is what makes a single evaluation sufficient to certify an entire envelope box, and nothing in the argument is pairwise—the common restriction of coupling analyses to hub pairs is not required.

B. Exact N -hub form of the omitted loss term

Exact DistFlow differs from (1) in that branch flows carry the losses generated downstream, plus a smaller $(r^2 + x^2)\ell$ term [11]. Writing $\mathcal{P}(m)$ for the root-to- m path and L_k for the loss flowing through branch k ,

$$D_{\text{err}}(m) = 2 \sum_{k \in \mathcal{P}(m)} r_k L_k, \quad L_k = \frac{1}{u} \sum_{\ell \in k} r_\ell S_\ell^2. \quad (3)$$

Theorem 2. Exchanging the order of summation in (3) gives the exact, non-pairwise identity

$$D_{\text{err}}(m) = \frac{2}{u} \sum_{\ell} r_\ell R_{m\ell} S_\ell^2, \quad (4)$$

where $R_{m\ell}$ is the resistance shared by bus m and branch ℓ . Substituting $S_\ell = B_\ell + \sum_h \mathbf{1}[\ell \in \mathcal{P}(h)] P_h$ yields

$$D_{\text{err}}(m) = \frac{1}{u} \left(P^\top C^{(m)} P + 2 b^{(m)\top} P + c^{(m)} \right), \quad (5)$$

$$C_{hh'}^{(m)} = \sum_{\ell \in \mathcal{P}(h) \cap \mathcal{P}(h')} r_\ell R_{m\ell}, \quad (6)$$

and $C^{(m)}$ is a Gram matrix of the vectors $(\sqrt{r_\ell R_{m\ell}} \mathbf{1}[\ell \in \mathcal{P}(h)])_e$, hence positive semidefinite: D_{err} is convex in the allocated caps.

Proof. A branch ℓ contributes to every $k \in \mathcal{P}(m)$ upstream of it, and the total resistance of those k is by definition $R_{m\ell}$; exchanging the two sums gives (4). Expanding the square gives (5), and $C^{(m)} = G^\top G$ with $G_{\ell h} = \sqrt{r_\ell R_{m\ell}} \mathbf{1}[\ell \in \mathcal{P}(h)]$ establishes positive semidefiniteness. $\square$

For two hubs A and B with $\rho = R_{AB}/R_{AA}$, (5) collapses to

$$D_{\text{err}}(A) \approx \frac{2R_{AA}^2}{V_{\text{ref}}^2} \left[\rho^2 S_{\text{sh}}^2 + (1 - \rho)^2 S_{\text{ex}}^2 \right], \quad (7)$$

where S_{sh} carries the sum of both hubs' draws through the shared segment and S_{ex} only hub A 's. Sharing therefore hurts superlinearly, and hubs on separate laterals do not interact even at high individual loading.

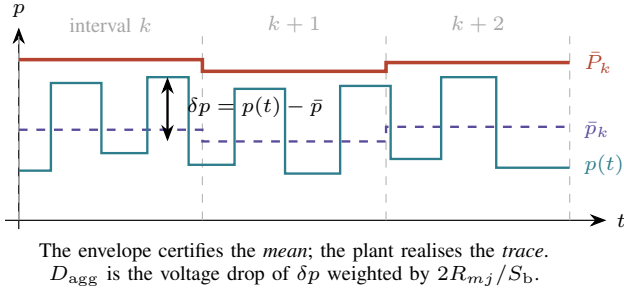


Fig. 3. Origin of the aggregation term. The envelope certifies the interval-mean injection; the plant realises a 1-minute trace inside the interval. D_{agg} is the voltage drop of that deviation, weighted by the network sensitivity.

C. Temporal aggregation

An envelope allocated once per interval of length Δ is planned against interval-mean injections $\bar{p}$, while the plant realises $p(t)$ (Fig. 3). Under (1) the resulting additional drop is exact and *first order*:

$$D_{\text{agg}}(m, t) = \frac{2}{S_b} \sum_j [R_{mj} \delta p_j(t) + X_{mj} \delta q_j(t)] \quad (8)$$

with $\delta p_j(t) = p_j(t) - \bar{p}_j$. If injection j drifts at bounded rate $|\dot{p}_j| \leq \gamma_j$ then $\max_t |\delta p_j| \leq \gamma_j \Delta/2$, giving the planning-time bound

$$D_{\text{agg}}(m) \leq \frac{\Delta}{2S_b} \sum_j [R_{mj} \gamma_j + X_{mj} \gamma_j^q]. \quad (9)$$

Collapsed onto two hubs with A at m , $D_{\text{agg}}(A) \leq (\Delta/2S_b) R_{AA} [\rho \gamma_{\text{sh}} + (1 - \rho) \gamma_{\text{ex}}]$. The structural contrast with (7) is the point: D_{err} is quadratic in power and carries ρ^2 , whereas D_{agg} is linear in drift and carries ρ^1 .

D. The criterion

The two terms are additive at first order, so

$$\Lambda_{\text{tot}}(m, t) = \frac{D_{\text{err}}(m, t) + D_{\text{agg}}(m, t)}{u^{\text{lin}}(m, \bar{p}(t)) - V_{\text{min}}^2} > 1 \quad (10)$$

predicts an AC violation at bus m . The denominator is the certified slack: the headroom the allocation's own linear model promised, which equals $V_{\text{plan}}^2 - V_{\text{min}}^2$ when the allocation binds. At a static operating point $D_{\text{agg}} \equiv 0$ and (10) reduces to the pure linearisation criterion. Where the realised trace is unavailable at planning time, the bound (9) replaces the second term of the numerator. No quantity in either term is fitted.

E. Where the mobility constant goes

τ does not appear in (10), and this is a result rather than an omission: the voltage constraint is instantaneous and electrical, so the *timing* of energy demand cannot enter it. τ governs energy feasibility—whether the fleet meets its departures—and it enters the voltage problem only through γ , by shaping how sharply hub power steps within an interval. As $\tau \rightarrow 0$ the hubs merge into one connection point: $R_{AB} \rightarrow R_{AA}$, $\rho \rightarrow$

1, $S_{\text{ex}} \rightarrow 0$, and (10) collapses to the stationary single-site condition

$$\frac{2R^2 S^2}{V_{\text{ref}}^2} + \frac{\Delta R \gamma}{S_b} > \text{band}. \quad (11)$$

Condition (11) is the ordinary stationary envelope safety condition, so the mobile case is a generalisation of the stationary distributed-energy-resource case, not a departure from it.

V. VALIDATION PROTOCOL

Two regimes are used. *Static*: an allocation is computed at one operating point, all hubs are pushed to their caps simultaneously (Theorem 1), and the corner is evaluated by AC sweep. *Closed loop*: a full day is simulated at 1-minute resolution with the envelope refreshed every 15 minutes and the hub MPC re-solved every 5 minutes; Λ_{tot} is evaluated at every minute at the bus the AC solution says binds.

Derivation set. 60 synthetic radial feeders with randomised branching and MV-plausible impedances, crossed with five ρ targets, three baseline margins and three planning bands.

Held-out feeders. The IEEE 33-bus and IEEE 69-bus systems, neither used in deriving or tuning anything. The 69-bus data are parsed directly from MATPOWER's `case69.m` [12], whose header cites Baran and Wu [11] and Das [13]; the parser is validated by reproducing the published base case, a minimum voltage of 0.9092 p.u. at bus 65, and 0.9131 p.u. at bus 18 for the 33-bus system, both at a 1.0 p.u. slack.

Closed-loop sets. Four topologies, 48 full-day runs, 69 120 simulated minutes. Sets A and B are the IEEE 33-bus system with hub C moved along a lateral through seven placements each; set B uses five placements never seen when the operating threshold was chosen, plus two at a larger fleet. Set C is the IEEE 69-bus system at a matched base-case minimum voltage (0.9567 against 0.9572). Sets D and E are two purpose-built synthetic feeders used for the manipulation experiment of Section VII-B.

VI. RESULTS

A. Static allocations

Table I reports the static regime. One-at-a-time allocation over-allocates by a median factor of exactly 2.000 for two hubs, matching the N -hub derivation, and violates in 98.4% of synthetic and 53–100% of published-feeder configurations. Jointly certified allocation with any band ≥ 0.005 p.u. violates in 0 of 1350 synthetic, 0 of 25 IEEE 33-bus and 0 of 25 IEEE 69-bus cases. The static classification is therefore degenerate—there are no positives to discriminate—so no receiver operating characteristic is reported and no threshold is fitted at a static point. On the one non-degenerate column (joint, zero band) the index exceeds unity for all 797 violations with 13 false positives.

Fig. 4 plots the resulting failure boundary: the band a planner must hold at a static operating point. Joint allocation requires 0.0055–0.0062 p.u., nearly flat in ρ ; one-at-a-time allocation requires 0.019–0.031 p.u. and rising.

TABLE I
STATIC REGIME. DEPTHS IN P.U.

Set	Mode	Band	n	Violated	Med. depth
Synthetic	one-at-a-time	0.000	810	98.4%	0.0186
Synthetic	one-at-a-time	0.005	810	98.4%	0.0111
Synthetic	one-at-a-time	0.010	810	57.2%	0.0019
Synthetic	joint	0.000	810	98.4%	0.0035
Synthetic	joint	0.005	810	0.0%	—
Synthetic	joint	0.010	540	0.0%	—
IEEE 33	one-at-a-time	0.005	15	100%	0.0107
IEEE 33	joint	0.005	15	0.0%	—
IEEE 69	one-at-a-time	0.005	15	53.3%	0.0127
IEEE 69	joint	0.005	15	0.0%	—

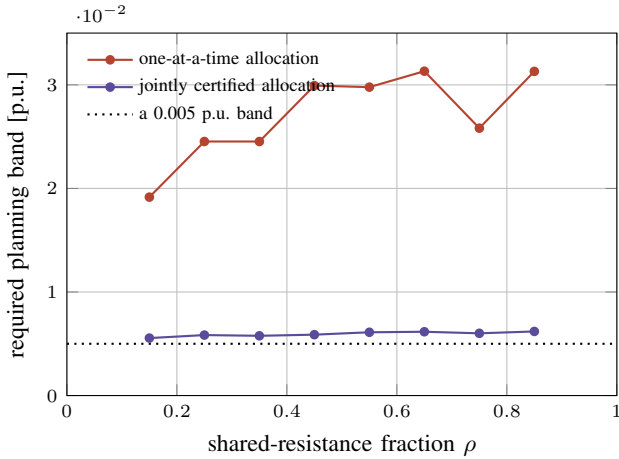


Fig. 4. Static failure boundary on the (ρ, margin) plane. Curves are the band required to stay secure; allocations below a curve are predicted to violate. These are static bands only—the closed loop adds up to a further 0.014 p.u.

B. N -hub check

Forty configurations were tested: the four-hub IEEE 33-bus case study and $H = 2, 3, 4, 5$ on IEEE 69-bus, at three baseline margins and three bands. The quadratic form (5) agrees with an independent loss-propagation computation to a maximum relative difference of 5×10^{-16} at every H ; $C^{(m)}$ in (6) is positive semidefinite in every case; and corner certification is never violated by the AC solver once a band exists.

C. Closed loop

Table II reports the closed-loop regime and Fig. 5 the violation counts. Adding D_{agg} raises per-minute recall from 5.6% to 82.2% at a 0.005 band and from 0% to 79.6% at 0.010, with zero false positives in all 69 120 simulated minutes.

Three placements at low coupling are the control that rules out a refresh-cadence-only explanation: at $\rho \leq 0.276$ the same cadence, fleet and drift magnitude produce *zero* violations, because the deviation lands at a bus with abundant certified slack. Cadence is necessary but not sufficient; coupling is what converts it into a violation.

TABLE II

CLOSED LOOP, BAND 0.005 P.U. SET A IS THE THRESHOLD-SELECTION SET; SETS B AND C WERE NOT USED TO CHOOSE ANY THRESHOLD

Set	ρ	Viol. min	D_{err}	D_{agg} p95	slack p5	TP/FN
A	0.008	0	0.00197	0.01046	0.01200	—
	0.086	0	0.00192	0.00655	0.01167	—
	0.276	0	0.00159	0.00398	0.01042	—
	0.515	51	0.00141	0.00979	0.00802	36/15
	0.696	69	0.00139	0.01292	0.00639	57/12
	0.817	79	0.00142	0.01357	0.00539	67/12
	0.934	87	0.00142	0.01398	0.00524	75/12
B	0.194	0	0.00176	0.00415	0.01131	—
	0.369	33	0.00143	0.00503	0.00942	0/33
	0.463	46	0.00141	0.00813	0.00850	24/22
	0.515 [†]	65	0.00161	0.00931	0.00574	46/19
	0.647	66	0.00140	0.01218	0.00682	51/15
	0.750	73	0.00142	0.01318	0.00595	62/11
	0.934 [†]	109	0.00179	0.00969	0.00159	96/13
C	0.000	0	0.00242	0.00910	0.00953	—
	0.250	0	0.00244	0.00872	0.01573	—
	0.465	0	0.00257	0.00881	0.01414	—
	0.611	0	0.00271	0.00756	0.01328	—
	0.731	27	0.00274	0.00737	0.01259	0/27
	0.860	42	0.00272	0.00782	0.01185	27/15

A, B: IEEE 33-bus; C: IEEE 69-bus. TP/FN at $\Lambda_{\text{tot}} > 1$. [†]20-vehicle fleet; all other rows 16 vehicles.

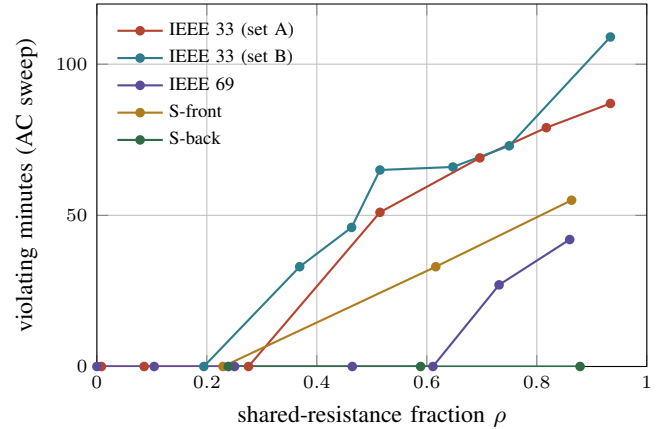


Fig. 5. Violating minutes against ρ , band 0.005 p.u., all closed-loop sets. The monotone rise and a violation-free low-coupling regime replicate on every topology; the coupling threshold at which violations begin differs ($\rho \approx 0.4$ on IEEE 33, $\rho \approx 0.7$ on IEEE 69), consistent with the latter's larger certified slack.

D. Threshold on a genuine holdout

The practical trigger $\Lambda_{\text{tot}} > 0.85$ was chosen on set A. Applied unchanged to the 28 runs of sets B and C, it preserves precision and false-positive rate exactly and loses recall (Table III). The value was not adjusted after seeing the holdout. It is therefore reported as bounded rather than universal: validated for $\rho \in [0.00, 0.93]$, bands 0.005 and 0.010 p.u., baseline margin ≈ 0.956 p.u., 15-minute refresh with 5-minute MPC re-solve, fleets of 16 and 20 vehicles, on two radial feeders. The derivation's own threshold remains $\Lambda_{\text{tot}} > 1$.

The recall loss is localised and explainable. The holdout

TABLE III
THRESHOLD GENERALISATION. HOLDOUT ROWS USE 0.85 EXACTLY AS
SELECTED ON SET A, WITH NO RETUNING

Set	Rule	Band	Recall	Precision	FPR
A (selection)	$\Lambda_{\text{tot}} > 1$	0.005	0.822	1.000	0.00000
A (selection)	$\Lambda_{\text{tot}} > 0.85$	0.005	0.955	1.000	0.00000
A (selection)	$\Lambda_{\text{tot}} > 0.85$	0.010	0.941	1.000	0.00000
B (holdout)	$\Lambda_{\text{tot}} > 1$	0.005	0.712	1.000	0.00000
B (holdout)	$\Lambda_{\text{tot}} > 0.85$	0.005	0.852	1.000	0.00000
B (holdout)	$\Lambda_{\text{tot}} > 0.85$	0.010	0.856	1.000	0.00000
C (holdout)	$\Lambda_{\text{tot}} > 1$	0.005	0.391	1.000	0.00000
C (holdout)	$\Lambda_{\text{tot}} > 0.85$	0.005	1.000	1.000	0.00000
C (holdout)	$\Lambda_{\text{tot}} > 0.85$	0.010	1.000	1.000	0.00000

placement at $\rho = 0.369$ sits at the lower edge of the violating regime, where violation depths are 0.0008–0.0010 p.u.—the same size as the first-order model’s own reconstruction residual, whose mean absolute error is 1.3×10^{-3} in u on IEEE 33 and 8.8×10^{-4} on IEEE 69. The index resolves violations deeper than its residual and cannot resolve shallower ones.

VII. WHAT DRIVES THE COUPLING DEPENDENCE

A. Decomposition

Measured as the 95th percentile of the aggregate D_{agg} , IEEE 33 shows a rise with ρ and IEEE 69 does not (Fig. 7). That statistic is misleading, and it is the statistic behind the apparent disagreement between sets in Section VI-C: the aggregate mixes the hub’s own drift term with a background from every other bus’s deviation, and a percentile of a signed sum is not the sum of its parts. Splitting the hub’s contribution into how much the fleet drifts and how much the network weights that drift,

$$D_{\text{agg}}^C(m) = w_C \gamma_C, \quad w_C = \frac{2R_{mC}}{S_b} = \frac{2R_{mm}}{S_b} \rho, \quad (12)$$

resolves it. Both factors in (12) were measured separately at every minute of all 48 closed-loop runs. The drift magnitude is invariant: γ_C is 90.0–91.1 kW (p95) in every run, at every ρ , on every feeder, at both fleet sizes and both bands. It is a property of the charger—90 kW is $0.6 \times$ hub C’s single 150 kW bay—not of the network. The entire ρ -dependence is w_C , which is proportional to ρ by construction: the correlation between ρ and w_C is +1.000 on every set. Hence

$$D_{\text{agg}}^C(m) = \frac{2R_{mm}\gamma_C}{S_b} \rho, \quad (13)$$

which contains no fitted quantity. Table IV and Fig. 6 compare it with measurement.

Four independent topologies and 22 placements agree with (13) to three significant figures. The one departure is reported rather than removed: the two 20-vehicle runs fall *below* the line, dragging that set’s fitted slope to 0.807 of the closed form. A larger fleet holds the hub nearer its cap for longer, so the bay switches less within an interval; (13) is an upper envelope in that regime rather than an equality.

TABLE IV
SLOPE OF THE HUB DRIFT TERM AGAINST ρ VERSUS THE CLOSED FORM
(13). BAND 0.005 P.U.

Set	R_{mm}	γ_C	obs.	$2R_{mm}\gamma_C/S_b$	ratio
A: IEEE 33	0.0690	90.1	0.01243	0.01243	1.000
B: IEEE 33, 16 veh	0.0690	90.0	0.01243	0.01242	1.000
C: IEEE 69	0.0463	90.2	0.00838	0.00836	1.003
D: synthetic, front	0.0474	90.1	0.00854	0.00854	1.000
E: synthetic, back	0.0474	90.1	0.00854	0.00854	1.000
B incl. 20-veh runs	0.0690	90.0	0.01002	0.01242	0.807

R_{mm} in p.u., γ_C in kW, slopes in p.u. of u per unit ρ .

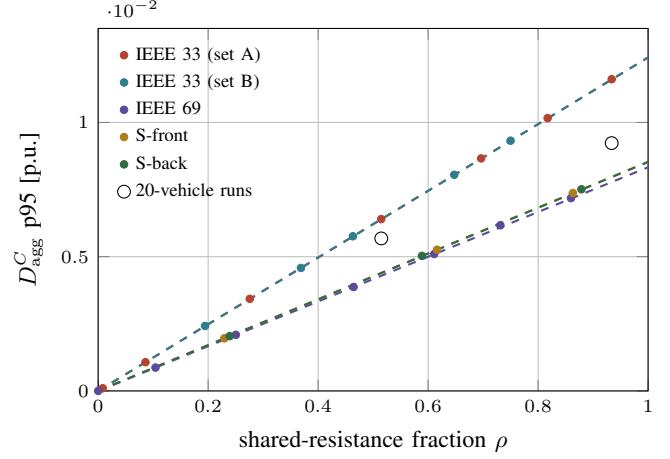


Fig. 6. The hub’s own drift term against ρ on four topologies and 22 placements. Dashed lines are the closed form (13) —predictions, not fits; markers are measured 95th-percentile values. Each set lies on its own line because R_{mm} differs; no slope is a free parameter. Circles mark the two 20-vehicle runs, the only points below their line.

TABLE V
MANIPULATION EXPERIMENT: ONE STRUCTURAL KNOB, MATCHED
BASE VOLTAGE, IDENTICAL ρ SWEEP INSIDE EACH ARM. BAND 0.005
P.U.

Arm	ρ	Viol. min	slack p5	D_{agg}^C p95	D_{err} med
front-loaded	0.229	0	0.01204	0.00196	0.00135
	0.616	33	0.00988	0.00526	0.00136
	0.863	55	0.00847	0.00737	0.00138
back-loaded	0.239	0	0.01761	0.00204	0.00156
	0.589	0	0.01619	0.00503	0.00165
	0.879	0	0.01501	0.00751	0.00164

B. Manipulation experiment

Correlation across two feeders is not a mechanism, so the structural hypothesis was tested by construction. Two synthetic radial feeders were built differing in exactly one property—whether the resistance of the hub’s path is concentrated near the substation (*front-loaded*) or accumulates toward the tip (*back-loaded*, the long thin lateral shape)—with total path resistance, hub layout, fleet, tariff and base-case minimum voltage (0.9570) all held equal, and the same three-point ρ sweep inside each arm (Table V).

The knob did not flip the mechanism: the drift term’s slope

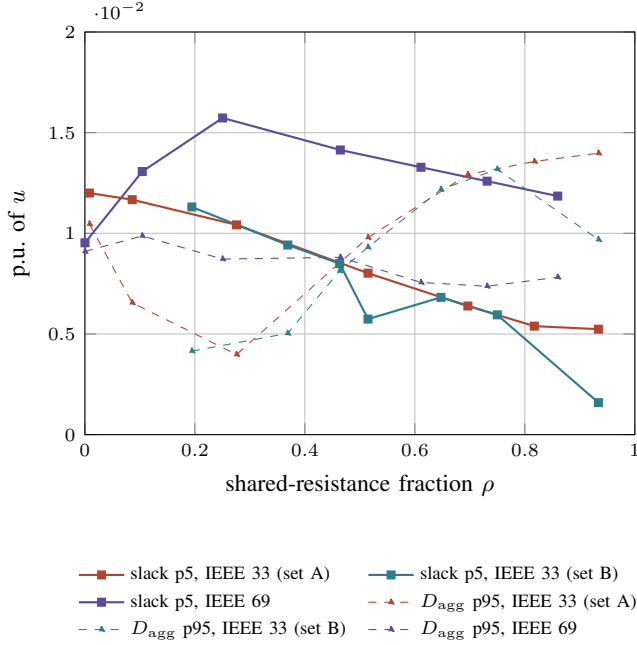


Fig. 7. Certified slack (solid, squares) and aggregate D_{agg} (dashed, triangles) against ρ on the three published-feeder sets. Slack falls with coupling on IEEE 33 and is flat on IEEE 69; the aggregate D_{agg} behaves inversely. Fig. 6 shows that once the hub's own term is isolated the two feeders agree exactly.

against ρ is identical in both arms (0.00854, the closed form to three figures), and slack falls with ρ in both. What the knob controlled is the slack *level* —0.0085–0.0120 front-loaded against 0.0150–0.0176 back-loaded at the same base voltage—and that decides whether anything violates at all: 0/33/55 violating minutes against 0/0/0, with the same fleet, the same ρ and the same drift. The supported causal structure is therefore resistance profile $\rightarrow$ slack level $\rightarrow$ violation incidence, separately from $\rho \rightarrow$ drift weight $\rightarrow D_{agg}$, with the second link exact and topology-independent. One confound is inherent and stated: to reach the same base margin the two arms carry different total load (0.415 against 1.150 of nominal), because a back-loaded profile drops less voltage per kilowatt of load.

C. One open question, narrowed

Certified slack falls with ρ on IEEE 33 (−0.0080, −0.0120) and in both synthetic arms (−0.0056, −0.0041) but is flat on IEEE 69 (+0.0007); Fig. 7 shows both quantities. The manipulation experiment rules out the resistance profile as the driver of that sign, since slack fell with ρ in both arms, and the invariance of γ_C rules out any fleet-drift explanation. What remains untested is the α -fair allocation's response on a feeder whose binding bus has a much longer upstream path, and the interaction between hub placement and the load composition of the other hubs. This is one sign of one derivative on one feeder; it does not affect Λ_{tot} , which uses the measured slack directly, and it is recorded as open rather than resolved.

VIII. GENERALISATION BEYOND BUS FLEETS

Nothing in the derivation refers to buses. The criterion requires three things: connection points that share feeder

resistance ($\rho > 0$), a load that can move between them, and an allocation computed per site and refreshed on an interval.

Delivery-fleet depots are the closest structural analogue, and the drift term is larger: depot power steps as vehicles plug in and out, and depots cluster on shared industrial laterals, so ρ runs high. The mobility constant is hours rather than minutes, which by Section IV-E changes γ but not the form of the criterion.

Mobile and relocatable chargers make ρ a decision variable rather than a network property: siting the unit chooses its shared resistance. Then $\Lambda_{tot} \leq 1$ is a siting constraint, not a band-derating rule—and because D_{agg} carries ρ^1 , siting buys more than de-rating.

Vehicle-to-grid and shared building storage make δp signed, which (8) handles exactly since it is linear. Reverse flow can raise voltage past the upper limit, so Theorem 1 must then be applied at the upper corner as well as the lower.

The portable statement is the parameterisation: under per-site envelopes, a fleet's grid impact is characterised by (ρ , band, max simultaneous draw, γ , Δ) and not by vehicle count, duty cycle or dwell time except through those five.

IX. THREATS TO VALIDITY

These are scope limits on what the evidence supports, not caveats on the derivation.

(a) *No measured feeder data.* Every closed-loop run applies synthetic load, photovoltaic and tariff profiles to standard test systems. Nothing here has been checked against metered distribution data, and real feeders carry load volatility, unbalance and tap-changer action this model does not represent. D_{agg} in particular is driven by intra-interval drift, exactly the quantity real data would be most likely to change.

(b) *One fleet model and one charger rating.* A single bus type, one route structure, one timetable generator and one charging technology. γ_C came out invariant at 90 kW, but that is $0.6\times$ the hub's single 150 kW bay: it is a charger property, and no run varied the bay rating or count, so the slope of (13) has been verified against topology but not against γ .

(c) *The operating threshold's generality is exactly what the holdout tested.* Two radial feeders, two bands, one refresh cadence and the ρ range of Section VI-D. The value 0.85 carries no warrant outside that box; $\Lambda_{tot} > 1$ is what should be used where it does not apply.

(d) *The index does not cover one-at-a-time allocation.* For that construction the index stays below unity while the AC solution violates, because it fails for a separate and larger reason—factor- N over-allocation—that neither error term models. The criterion is valid for jointly certified envelopes and under-predicts otherwise.

X. CONCLUSION

Per-site dynamic operating envelopes are secure in the model that produces them, and this paper gives the condition under which that security survives in the physical network when the flexible load is a mobile fleet sharing feeder resistance. Two additive error terms are derived from the branch-flow equations: an omitted-loss term, exact at N hubs and

convex in the allocated caps, and a temporal-aggregation term that is first-order in intra-interval drift. Their sum normalised by the certified slack gives one checkable inequality, $\Lambda_{\text{tot}} > 1$ in (10). That threshold is the general claim of this work: it follows from the derivation, contains no fitted quantity, and held with perfect precision and zero false positives across all 48 closed-loop runs and both held-out published feeders. The relaxed trigger $\Lambda_{\text{tot}} > 0.85$ is a secondary, operational recommendation only: it buys recall inside the tested box of Section VI-D and, as Section IX states, carries no warrant outside it.

The empirical contribution is that the aggregation term is what makes envelope failure a coupling phenomenon rather than a refresh-rate one, and that its coupling dependence is entirely the network sensitivity weight: fleet drift magnitude was invariant across every run, giving the closed form (13), which matched measurement to three significant figures on four topologies without a fitted quantity. A manipulation experiment separated what controls violation incidence—the resistance profile, through slack level—from what controls the mechanism—coupling, through the weight. Future work should replace the synthetic profiles with measured feeder data, vary the charger rating to test the closed form against γ as well as topology, and resolve the open sign reported in Section VII-C.

REPRODUCIBILITY

All simulation code, the topology generators, the parsed test-system data and the result files behind every table and figure are available from the authors. Every reported violation is produced by the AC solver at every simulated minute, never by the linear model an allocation was optimised against.

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Author Two Biography text here.